

Lesson Exemplar in General Science

Quarter 3

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9

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Lesson Exemplar for General Science
Quarter 3: Lesson 9

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LESSON EXEMPLAR

Learning Area	GENERAL SCIENCE	Grade Level	11
Semester	SECOND	Quarter	3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	<i>The learners learn that climate change affects ecosystems and the organisms living in them.</i>
Performance Standard	<i>The learners use examples to describe the relationships among living systems and their components, as well as how the structure of a part related to its function. They explain the role of cells in the perpetuation of life and how organ systems work together to maintain good health. They conduct investigations on the effects of climate change in local ecosystems.</i>
Learning Competencies	<i>In this lesson, the learners will be able to:</i> <i>describe how biological systems respond to climate change and contribute to understanding its effects on living organisms in the Philippines;</i>

II. REFERENCES and MATERIALS

(Selecting Resources and Material)

References:

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Panela, Shaira. "Too Many People': Philippine Island Being Deforested despite Extensive Protections." *Mongabay Environmental News*, October 31, 2014. <https://news.mongabay.com/2014/10/too-many-people-philippine-island-being-deforested-despite-extensive-protections/>.

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III. CONTENT <i>(Sequencing Content)</i>	<i>Climate Change and Biological Systems</i>
IV. OBJECTIVES	• Identify <i>changes in biological systems (e.g., animal migration, coral bleaching, forest dieback) due to climate change.</i>

(Setting Clear Objectives and Analyzing the Tasks)	• Explain how organisms adapt or fail to adapt to climate-related stressors (e.g., drought, flooding, temperature rise). • Relate scientific evidence to observable effects in Philippine ecosystems (e.g., mangrove loss, decline in rice yield, species endangerment). • Analyze how these biological responses help scientists and communities understand the broader impact of climate change. • Reflect on how climate change affects biodiversity, and food security in the Philippines.					
IV. PROCEDURES (Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)		ANNOTATION				
A. Activating Prior Knowledge	1. Activating Prior Knowledge Option 1: Knowledge Check! Recall your ideas about “Climate Change and Biological Ecosystems”. Write as many related words as you can within a period of five minutes. Write your answers on the corresponding box below.	This strategy tap into learners’ existing knowledge , encouraging metacognition. These activities also promote learner voice as they define terms using their own language and helps bridge the gap between scientific terms and lived experiences.				
	<table><tr><th>Climate Change</th><th>Biological Ecosystems</th></tr><tr><td>Example: Burning of fossil fuels</td><td>Example: Forest Ecosystems</td></tr></table>		Climate Change	Biological Ecosystems	Example: Burning of fossil fuels	Example: Forest Ecosystems
	Climate Change		Biological Ecosystems			
Example: Burning of fossil fuels	Example: Forest Ecosystems					

Option 2: Word Wall Warm-up

Activate your vocabulary related to climate change and biological ecosystems. On the board or wall, key terms are posted.

- ✓ *Select 5 terms you know well.*
- ✓ *Define them in their own words.*
- ✓ *Discuss how each might relate to climate or ecosystems.*

Possible word wall:

WORD WALL			
biotic	forest	coral reef	mangrove
freshwater	rainforest	desert	abiotic
producers	biodiversity	habitat	population
greenhouse gases		sea level rise	
drought	coral bleaching		migration
crop failure		mitigation	

Option 3. Four Corners – Climate & Ecosystem Beliefs

Label four corners of the room with:

- ✓ **Strongly Agree**
- ✓ **Agree**

*This activity engages learners' existing knowledge and helps expand their vocabulary about climate change and biological ecosystems. By selecting and explaining terms in their own words, learners deepen their understanding through **personal relevance**. Talking about how each term connects to climate or ecosystems fosters **critical thinking** and enhances learning within a meaningful context.*

*The interactive nature of the Four Corners activity supports **learner-centered** and **contextualized learning**, empowering learners to connect their beliefs with scientific ideas in a respectful and open environment. The flexibility to adapt the activity using hand*

	✓ Disagree ✓ Strongly Disagree The teacher will read the following statements: ✓ "Climate change is caused only by natural processes." ✓ "All ecosystems can recover from damage." ✓ "Humans are the main drivers of biodiversity loss." ✓ "Coral reefs are not important to humans." Then, the learners move to the corner that matches their opinion and explain why. Note: The teacher could modify this activity to what fits the learners. He/She could either ask learners to have corresponding hand signals to each of the choices. 2. Establishing the Purpose of the Lesson Option 1: Mystery Photo Prompt Show learners a set of pictures or short video clips like: ✓ Bleached coral reef ✓ Dying forest or wildfire aftermath	signals or movement addresses inclusivity and accommodates diverse learner needs. The use of guided questions encourages inquiry-based learning and communication skills, as learners articulate their observations and hypotheses, fostering deeper understanding and environmental awareness. This method also cultivates values formation, helping learners recognize human impacts on the planet and the urgency of climate action.
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- ✓ Animal species in an unusual habitat (e.g., fish in flooded fields)
- ✓ Dry, cracked farmlands

The teacher may use the following pictures:



*By linking visual evidence to climate concepts, the activity aligns with the **learner centered** approach of the MATATAG curriculum, making learning relevant and meaningful. It further encourages **collaboration** when done in pairs or groups, reinforcing social interaction and shared knowledge construction.*



	Guide questions: 1. What do you see? 2. What might have caused these changes? 3. How do you think this connects to climate? Option 2: What If...?” Scenario Cards The teacher introduces climate stressors and predicts biological responses by giving learners “What If” scenario cards. The following could be given to the learners: ✓ “What if the temperature in a coral reef rose by 2°C?” ✓ “What if typhoons became more frequent in a coastal mangrove area?” ✓ “What if drought lasted for 6 months in a farming region?” Guide questions: 1. What might happen to plants and animals? 2. How might humans be affected? 3. Can the system recover?	<i>This activity leverages skills on critical thinking and problem-solving as they explore the impacts of climate stressors on biological systems through hypothetical scenarios. By considering situations such as increased temperatures in coral reefs, more frequent typhoons in mangrove areas, and prolonged droughts in farming regions, learners develop an understanding of ecosystem vulnerabilities and resilience.</i> <i>The guiding questions promote inquiry-based learning, encouraging learners to analyze the interconnected effects on plants, animals, and human communities, and to evaluate the potential for ecosystem recovery.</i> <i>Through discussion and collaboration, learners practice communication skills and values formation, reinforcing their role as responsible stewards of the environment.</i> <i>This activity fosters higher-order thinking and systems-based reasoning by guiding learners to analyze how climate</i>
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	Option 3: Living Systems Under Pressure Learners choose or are assigned an ecosystem like forest, coral reef, rice field. Then, they will answer the following questions: 1. What does such ecosystem look like in a healthy state? 2. What happens when a climate-related event (heat, flood, etc.) occurs? 3. How humans could help restore it?	stressors influence biological systems. By responding to open-ended questions, learners move beyond surface observations to consider complex biological reactions, the capacity of systems to adapt, and the likelihood of recovery. This approach supports the focus on cultivating scientifically literate learners who can evaluate environmental challenges and think critically about interconnected systems.																		
B. Instituting New Knowledge	3. Developing Mastery Option 1. Link the Ideas! Complete the table below by matching the impact of climate change to how biological systems respond through them. Select your answers from the boxes and write them on the corresponding rows and columns. <table><tr><th>Climate Change Impact</th><th>Response of Biological Systems</th></tr><tr><td>1.</td><td>1.</td></tr><tr><td>2.</td><td>2.</td></tr><tr><td>3.</td><td>3.</td></tr><tr><td>4.</td><td>4.</td></tr><tr><td>5.</td><td>5.</td></tr><tr><td>6.</td><td>6.</td></tr><tr><td>7.</td><td>7.</td></tr><tr><td>8.</td><td>8.</td></tr></table>	Climate Change Impact	Response of Biological Systems	1.	1.	2.	2.	3.	3.	4.	4.	5.	5.	6.	6.	7.	7.	8.	8.	This activity supports learners in making meaningful connections between climate change impacts and the responses of biological systems, encouraging analytical thinking and knowledge integration. By matching each environmental change with an appropriate biological reaction, learners deepen their understanding of cause-and-effect relationships in ecosystems. The matching format promotes active learning, allowing learners to work independently or collaboratively. It also integrates science and literacy, as learners must comprehend scientific terms and select context-appropriate responses. When processed through class discussions or group
Climate Change Impact	Response of Biological Systems																			
1.	1.																			
2.	2.																			
3.	3.																			
4.	4.																			
5.	5.																			
6.	6.																			
7.	7.																			
8.	8.																			

9.	9.
10.	10.

*sharing, this activity can build communication skills and promote **values formation** by fostering empathy and responsibility toward the environment.*

Climate Change Impact

- Rising global temperatures
- Ocean warming
- Deforestation
- Prolonged droughts
- Sea level rise
- Melting polar ice caps
- Frequent typhoons
- Habitat destruction
- Flooding of farmlands
- Greenhouse gas emissions

Responses of Biological Ecosystems

- Coral bleaching
- Loss of biodiversity and extinction
- Migration of animals to cooler areas
- Coastal mangrove forest damage
- Crop failure and food insecurity
- Habitat loss for polar bears and seals
- Increased flooding and erosion in lowland areas
- Disruption of marine ecosystems
- Carbon dioxide buildup from burning fossil fuels
- Reduced rainfall affecting freshwater ecosystems

	Option 2. Survival of the Fittest – Climate Edition Explain how organisms adapt or fail to adapt to climate-related stressors. Listed on BOX 1 are different organisms and climate-related stressors on BOX 2. Choose one organism and stressor from the boxes and identify how each organism responds and adapt to such climate-related stressor. BOX 1 rice bangus tarsier mangrove tree BOX 2 drought sea-level rise intense typhoons prolonged heat	<i>This part of the lesson makes the curriculum relevant to the learners. Activities use real-world and local examples like mangroves, bangus, tarsier, etc.</i>
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Note: This activity could be done by group. Each group presents their organism's fate and adaptive strategies or failure in adapting such climate-related stressors.

Option 3: Ecosystem Picture Sort

Provide learners with a set of images or cards showing different ecosystems like rainforest, coral reef, mangrove, mountain, grassland. Learners sort them into types (terrestrial, aquatic, coastal). Then, they will discuss what plants, animals, and weather are found there and predict how each might be affected by climate change.

The teacher may use the following pictures:



*This activity supports **holistic development** by integrating **science, literacy, and socio-emotional learning**, and helps build a strong foundation for **climate action and resilience**, crucial in preparing learners to become **environmentally responsible Filipino citizens**.*



C. Demonstrating Knowledge and Skills	1. Finding Practical Application Option 1: Let's Watch! Present a 3-5-minute clip about the dramatic biological changes due to climate events. Let the learners respond with: • One word to describe what they felt or saw • One question they now have Then, the learners will share their insights through pairs or by group. A sample video could be accessed through the following link: https://www.youtube.com/watch?v=Ev2M3T21YrU	<i>This activity uses a short 3–5-minute video clip to visually highlight the dramatic biological changes caused by climate events. After viewing, learners will reflect by sharing one word that captures their emotion or observation and one question they now have. This encourages emotional engagement and curiosity. Learners will then discuss their insights in pairs or small groups, promoting collaboration, critical thinking, and awareness of climate change's impact on living systems.</i>

	Option 2: Essay Write a short essay (150–200 words) in any of the following question: 1. "Why should Filipinos care about how climate change affects our ecosystems and biodiversity?" 2. "How does climate change threaten biodiversity?" Option 3: Interview Local Experts or Community Members Gather first-hand insights on how climate change affects your community through an interview. You can interview a local farmer, fisherman, barangay official, environmental workers or environmental advocates in your locality. Summarize the results of your interview and present it to the class. Note: The teacher will let the learners prepare their own interview questions related to climate change and how biological responds through them. Highlight also how the scenario/s affect the people in	The purpose of this task is to develop learners' critical thinking and written communication skills while deepening their understanding of the relationship between climate change and biodiversity loss. By choosing a topic that connects scientific concepts to real-life issues, especially within the Philippine context, learners are encouraged to reflect on environmental responsibility and the urgency of protecting ecosystems. This activity also provides an opportunity for personal expression and informed opinion, reinforcing both academic and civic awareness. Utilizing this activity directly aligns with constructivist learning. By engaging in problem-solving, collaboration, and personal meaning-making, learners construct new knowledge that is relevant, contextualized, and applicable to their community and environment.
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	the community and what actions do they undertake to address such impacts. 2. Making Generalization Option 1: “3-2-1” Reflection Sheet Write: • 3 things they learned about how climate change affects biological systems • 2 specific examples from the Philippines • 1 thing they can do to help reduce climate change impacts Option 2: One-Sentence Summary Challenge "In one sentence, explain how climate change affects biological systems and why it matters." Option 3: Think-Pair-Share: Big Takeaways Learners will be asked to accomplish the following tasks: • Think: Write an individual response on the question • Pair: Discuss with a partner	This reflective activity engages learners in summarizing and personalizing their learning on how climate change affects biological systems. It promotes comprehension, local relevance, and personal responsibility. It encourages learners to process key information, connect it to real-life situations in the Philippines, and think critically about their role in addressing climate issues. By summarizing complex ideas in one sentence, learners develop the skill of expressing important scientific insights in a simple yet meaningful way. This activity uses the Think-Pair-Share strategy to encourage deep reflection and collaborative learning. They then pair up to discuss their thoughts, compare ideas, and gain new perspectives. Finally, they share their insights with the whole class or in small groups. This structured approach promotes active participation, critical thinking, and peer learning, while helping learners
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	• <i>Share: Share with the class or in small groups</i> Sample Question: <i>“How does understanding biological systems help us see the bigger picture of climate change?”</i> 3. Additional Activities Option 1: Climate Change Case Study Learners will explore how climate change affects local biological systems by observing their surroundings and gathering information from community members such as farmers and fisherfolk. Their presentations will showcase observed changes and how people in their area are adapting, helping connect classroom learning to real-life situations. Use the following format: 1. Title of the Case Study 2. Introduction - Brief description of the community or area studied. - Purpose of the case study (to observe/document biological changes linked to climate change) 3. Phenomenon Observed - Description of the observed change (e.g., coral bleaching, pest outbreak, changes in bird migration) - (Include photo/drawing if available) 4. Connection to Climate Change - Explain how the phenomenon is related to climate change (temperature rise, irregular rainfall, stronger typhoons, etc.)	<i>connect scientific concepts to real-world climate issues.</i> <i>Using Philippine ecosystems and community concerns to make learning meaningful targets contextualization of the content of the lesson.</i> <i>Also, learners collaborate, analyze data, create action plans, and communicate solutions. Thus, catering the idea of the advancement of the 21st Century Skills.</i>
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	5: Effects on the Ecosystem - Impact on plants, animals, or biodiversity - Changes in productivity or availability of resources 6: Effects on the Community -Impact on livelihoods (farmers, fisherfolk, families) -Social or economic challenges faced by people 7: Adaptations or Responses -How the community is coping (new practices, adjustments, innovations) - Possible solutions learners recommend 8. Conclusion - Key insights about the connection between climate change, ecosystems, and community life 9. Proposed Solution - Plan of action to address the problem. Reflection and Discussion Questions: 1. What college courses or professions are directly related to your case study? 2. What skills did you develop while doing the case study (e.g., research, interviewing, data gathering, analysis, teamwork)? 3. Do you see yourself pursuing a career that addresses environmental or community concerns? Why or why not? 4. How might jobs related to biodiversity, agriculture, or environmental science become more important in the future? Option 2: Ecosystem Evidence Wall (Gallery Walk)	<i>Working in rotating groups, learners connect scientific evidence with real-world environmental issues, enhancing their analytical and observational skills. The activity encourages collaboration, critical</i>
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	Relate scientific evidence to observable effects in Philippine ecosystems. Look and observe the different learning stations around the room. <i>Possible learning stations:</i> ✓ Mangrove loss in Palawan ✓ Decline in rice yield in Central Luzon ✓ Philippine Eagle habitat loss ✓ Coral reef degradation in Tubbataha ✓ Drought impact in Mindanao Leaners rotate in groups and answer guided questions: 1. What is happening here? 2. What scientific data is shown? 3. How does climate change play a role? Option 3: Reflection Journal or Podcast – Voices for Biodiversity Reflect on how climate change affects biodiversity and food security in the Philippines. Learners choose between: • Writing a journal entry from the perspective of a Filipino farmer, fisherfolk, or wildlife researcher • Recording a short podcast episode (2–3 minutes) explaining: 1. What biodiversity loss they see/feel 2. How food security is being threatened 3. What they hope for the future	<i>thinking, and a deeper understanding of how climate change directly affects biodiversity and ecosystems in the Philippines.</i> <i>This activity promotes empathy, critical thinking, and awareness of local environmental issues while enhancing learners’ communication skills. By engaging with real-world challenges through a personal lens, learners connect scientific concepts with social realities, deepening their understanding of how climate change affects both ecosystems and human lives in the Philippines.</i>
V. ASSESSMENT (Assessing Learnings)	Option 1: Multiple Choice <i>Instructions: Choose the letter of the correct answer.</i>	

1. Which of the following is the primary cause of coral bleaching?
A. Oil spills B. Overfishing C. Rising ocean temperatures D. Lack of sunlight
2. Which mainly triggers the forest dieback in tropical rainforests?
A. Invasive species B. Drought and extreme heat C. Excessive rainfall D. Logging alone
3. Which of the following **BEST** describes adaptation in biological systems?
A. Migrating to other places
B. Dying from environmental changes
C. A random mutation that causes disease
D. A long-term genetic change that helps an organism survive
4. What is an example of an **ecosystem affected by climate change in the Philippines**?
A. Urban highways B. Mangrove forests C. Cemented rice terraces D. Underground caves
5. Which of the following is an example of an organism failing to adapt to climate change?
A. Migratory birds shifting their timing
B. Coral reefs bleaching and dying
C. Rice plants growing with less water
D. Trees growing at higher elevations
6. Which Philippine ecosystem is **MOST** threatened by rising sea levels?
A. Upland forests B. Grasslands C. Mangrove forests D. Mountain caves
7. What might happen if rice paddies experience more frequent flooding due to typhoons?
A. Increased rice production
B. Improved irrigation
C. Decline in rice yield
D. More food security
8. Which is an example of a climate-related biological response?
A. Photosynthesis
B. Migration of animals to cooler regions

- C. Use of fertilizers
- D. Seasonal planting of crops

9. Which species is most likely to migrate in response to climate change?
 A. Coconut tree B. Whale shark C. Mushroom D. Rice plant

10. Which of the following **BEST** describes biodiversity?
 A. The number of humans on Earth
 B. The amount of pollution in the environment
 C. The variety of living organisms in an area
 D. The size of an ecosystem

Option 2: Matching-Type: Climate Change and Biological Ecosystems

*Instructions: Match the items in **Column A** (Climate Change Impact) with the correct item in **Column B** (Response of Biological Systems). Write the letter of the correct answer before each number.*

Column A – Climate Change Impact

Column B – Response of Biological Systems

1. Rising global temperatures
2. Ocean warming
3. Deforestation
4. Prolonged droughts
5. Sea level rise
6. Melting polar ice caps
7. Frequent typhoons
8. Habitat destruction
9. Flooding of farmlands

- A. Coral bleaching
- B. Loss of biodiversity and extinction
- C. Migration of animals to cooler areas
- D. Coastal mangrove forest damage
- E. Crop failure and food insecurity
- F. Habitat loss for polar bears and seals
- G. Increased flooding and erosion in lowland areas
- H. Disruption of marine ecosystems
- I. Carbon dioxide buildup from burning fossil fuels

APPENDIX

Answer Key

Assessment

Option 1: Multiple Choice

Correct Answer	Rationalization
1. C. Rising ocean temperatures	Coral bleaching occurs when corals expel the symbiotic algae (zooxanthellae) living in their tissues due to stress from elevated sea temperatures, which is mainly driven by climate change. This leaves the coral white (bleached) and vulnerable.
2. B. Drought and extreme heat	Forest dieback is often caused by prolonged drought and rising temperatures, which weaken trees and disrupt ecosystem functions, leading to large-scale tree mortality.
3. D. A long-term genetic change that helps an organism survive	Adaptation in biology refers to inherited traits or genetic changes that improve an organism's chance of survival and reproduction in a changing environment over time.
4. B. Mangrove forests	Mangrove ecosystems in the Philippines are threatened by rising sea levels, changes in salinity, and increased storm surges due to climate change.
5. B. Coral reefs bleaching and dying	When coral reefs bleach and die due to temperature rise, it indicates their inability to adapt quickly enough to the rapidly changing climate.
6. C. Mangrove forests	Mangrove forests are located in coastal zones and are directly affected by sea-level rise, which can submerge their root systems and alter salinity levels.
7. C. Decline in rice yield	Flooding can damage rice crops, especially during critical growth stages, leading to reduced productivity and threatening food security.
8. B. Migration of animals to cooler regions	As temperatures rise, many animals move to cooler areas to survive. This shift in distribution is a direct biological response to climate change.

9. B. Whale shark	Whale sharks, being mobile marine animals, can migrate to new areas in search of optimal temperatures or food, unlike plants and fungi which are immobile.
10. C. The variety of living organisms in an area	Biodiversity refers to the richness and variety of life—plants, animals, fungi, and microorganisms—in a particular habitat or on Earth as a whole.

Option 2: Matching Type

Correct Answer	Rationalization
1. Rising global temperatures – C. Migration of animals to cooler areas	Animals shift their range toward higher altitudes or latitudes to escape the heat.
2. Ocean warming – A. Coral bleaching	Warmer ocean temperatures stress corals, causing them to expel algae and turn white (bleach).
3. Deforestation – B. Loss of biodiversity and extinction	Cutting down forests destroys habitats, threatening countless species with extinction.
4. Prolonged droughts – J. Reduced rainfall affecting freshwater ecosystems	Droughts decrease water availability, harming rivers, wetlands, and aquatic life.
5. Sea level rise – D. Coastal mangrove forest damage	Mangroves are threatened by submersion and saltwater intrusion due to rising seas.
6. Melting polar ice caps – F. Habitat loss for polar bears and seals	Ice-dependent species lose their homes and hunting grounds as ice melts.
7. Frequent typhoons – G. Increased flooding and erosion in lowland areas	Stronger storms bring heavy rains that flood and erode vulnerable regions.
8. Habitat destruction – H. Disruption of marine ecosystems	Coastal development and pollution destroy breeding and feeding areas for marine life.
9. Flooding of farmlands – E. Crop failure and food insecurity	Floods ruin crops, reduce harvests, and threaten local and national food supply.
10. Greenhouse gas emissions – I. Carbon dioxide buildup from burning fossil fuels	Emissions from cars, factories, and power plants increase atmospheric CO ₂ , driving global warming.

Option 3: Short Answer

1. How does rising temperature affect coral reefs in the Philippines?

- Rising ocean temperatures stress coral reefs, causing them to expel the symbiotic algae (zooxanthellae) that live inside their tissues. This leads to coral bleaching, where corals turn white and become more vulnerable to disease and death. In the Philippines, which has some of the most diverse coral reefs in the world, this results in loss of marine biodiversity, damages to fisheries, and reduced protection for coastlines from waves and storms.

2. Describe how drought caused by climate change can affect agriculture.

-Drought reduces the availability of water needed for crops to grow. This can lead to lower crop yields, drying of soil, and increased risk of crop failure, especially in water-dependent farming systems like rice cultivation. In the Philippines, prolonged droughts threaten food security, raise food prices, and harm the livelihood of farmers who rely on consistent rainfall.

3. What is forest dieback, and why is it happening more often today?

-Forest dieback is the gradual dying of trees in a forest, often over large areas. It happens when trees become too stressed by environmental changes, such as extreme heat, prolonged drought, pests, or wildfires. Today, it is more common because climate change increases these stressors, weakening tree health and disrupting forest ecosystems.

4. How do mangroves help protect coastal ecosystems and communities?

-Mangroves act as natural barriers that absorb wave energy, reducing coastal erosion and protecting communities from storm surges and flooding. They also provide nursery habitats for fish and marine life, help trap sediments, and store carbon, making them crucial for both biodiversity and climate resilience.

5. Why are animals migrating to new areas due to climate change? Give one example.

-As temperatures rise and habitats change, animals migrate to find cooler, more suitable environments for survival. This is a biological response to avoid heat stress, find food, or reproduce. For example, some bird species in the Philippines are shifting their range to higher elevations in mountainous areas to escape the rising temperatures in the lowlands.